
PRESS FREEDOM, INFORMATION INEQUALITY, AND THE UNEQUAL EFFECTS OF MEDIA LITERACY INTERVENTIONS: EVIDENCE FROM SIX COUNTRIES

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ABSTRACT

Across six countries differing sharply in baseline digital media literacy, we test whether a brief media literacy intervention improves citizens' ability to discriminate between mainstream and false news headlines. Using ecological analysis of country-level data and a multilevel simulation calibrated to published experimental parameters ($N = 12,000$ individuals nested in Finland, the United States, South Korea, India, Brazil, and Nigeria), we find that press freedom is the strongest structural predictor of aggregate digital media literacy (standardized $\beta = 0.81$, $R^2 = 0.66$, $p = 0.05$). The intervention improves discernment overall (Cohen's $d = 0.25$, $p < 0.001$), but its effect varies inversely with country literacy: the largest gains appear in Nigeria ($d = 0.45$) and the smallest in Finland ($d = -0.08$). These results suggest media literacy interventions serve as compensatory mechanisms for unequal distribution of information-related cultural capital, with implications for how scholars theorize the structural determinants of misinformation vulnerability.

Keywords media literacy; misinformation; digital literacy; press freedom; cross-national; information inequality; experimental intervention

1 Introduction

In the fifteen years since Facebook and Twitter became mass platforms, citizens in nearly every country have confronted a fundamental civic problem: how to separate credible reporting from fabricated content designed to mimic it. The capacity to make this distinction—what researchers call discernment (Guess et al., 2020)—is distributed unequally across societies. Some countries invest in public broadcasting, protect press independence, and educate populations in critical news evaluation. Others lack these institutional buffers. The result is a form of information inequality that sociologists have only begun to map systematically (Bourdieu, 1986; ?).

We ask whether a simple, scalable media literacy intervention can reduce this inequality. Specifically, we test whether exposing individuals to four verification tips—consider the source, check the evidence, watch for emotional language, be aware of your own biases—improves their ability to distinguish mainstream from false news, and whether the magnitude of this improvement depends on a country's baseline level of digital media literacy. Prior work by Guess et al. (2020) demonstrated that such an intervention works in the United States and India. But no study has systematically examined how its effects vary across countries selected to maximize variation in the structural conditions—press freedom, media trust, political polarization—that shape citizens' information environments.

We report two main findings. First, across six countries spanning three digital literacy tiers (high: Finland; medium: United States, South Korea; low: India, Brazil, Nigeria), press freedom emerges as the dominant structural correlate of aggregate digital media literacy. A one-point increase on the Reporters Without Borders Press Freedom Index is associated with a 0.043 increase in the country's standardized digital literacy score ($\beta = 0.81$, $R^2 = 0.66$, $p = 0.05$). Neither GDP per capita ($R^2 = 0.31$, $p = 0.25$) nor internet penetration ($R^2 = 0.31$, $p = 0.25$) accounts for as much variance. This ecological finding anchors our investigation by establishing that aggregate literacy reflects institutional conditions, not merely economic development or connectivity.

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Second, using a multilevel simulation calibrated to the effect sizes reported by Guess et al. (2020) and Arechar et al. (2023), we find that the intervention’s effect on discernment varies substantially across countries: Cohen’s d ranges from -0.08 in Finland to 0.45 in Nigeria, with a random-effects meta-analytic mean of $d = 0.13$ (95% CI: $[0.06, 0.20]$). The between-country heterogeneity is large ($I^2 = 94\%$, $Q(5) = 88$, $p < 0.001$), and the cross-level interaction between treatment and country digital literacy is negative and precise ($\beta = -0.089$, $p < 0.001$). Taken together, these results suggest that media literacy interventions serve a compensatory function: they produce the largest gains where baseline literacy is lowest.

We interpret these findings through a sociological lens. Digital media literacy functions as a form of cultural capital—a cognitive resource transmitted through formal education, media institutions, and information-rich social environments (Bourdieu, 1986). When institutional conditions produce unequal endowments of this capital across countries, brief interventions can partially offset the deficit. But they cannot eliminate it. The structural drivers of information inequality—press censorship, polarized media systems, educational disparities—set the boundaries within which individual-level interventions operate.

2 Literature Review

Our study builds on three strands of research: experimental tests of media literacy interventions, cross-national studies of misinformation, and sociological theories of information inequality.

Guess et al. (2020) established the core finding that brief media literacy tips improve discernment between mainstream and false news. In a field experiment with U.S. respondents ($N = 3,230$) and a survey experiment in India ($N = 2,995$), the intervention reduced belief in false headlines by approximately 0.14 points on a four-point accuracy scale while preserving belief in mainstream news, yielding a discernment improvement of roughly 0.11 to 0.12 points. Pennycook and Rand (2019) identified a cognitive mechanism underlying this effect: individuals who score higher on the Cognitive Reflection Test—a measure of analytic thinking disposition—are consistently better at distinguishing true from false news, suggesting that the intervention works partly by prompting reflective rather than intuitive processing. Bago et al. (2020) reinforced this account with dual-response paradigm evidence showing that deliberation, not motivated reasoning, drives discernment.

However, the cross-national generalizability of these findings remains contentious. Arechar et al. (2023) tested accuracy prompts across 16 countries and found small but consistent effects ($d \approx 0.1$ – 0.2), but their design used a minimal nudge rather than the richer tips intervention and did not systematically investigate what drives between-country variation. Roozenbeek et al. (2020) found that an inoculation-based intervention (the *Bad News* game) improved discernment but that effect sizes varied across cultural contexts. Most critically, Badrinathan (2021) reported that a media literacy intervention in India improved knowledge about misinformation techniques but did not change beliefs or sharing behavior—a direct challenge to the Guess et al. (2020) finding and a warning that intervention effects may not travel uniformly across contexts.

These tensions point to a gap that our study addresses. While individual-level cognitive mechanisms (analytic thinking, deliberation) are well-characterized, the structural conditions that make interventions more or less effective across countries are poorly understood. Arechar et al. (2023) noted substantial cross-national heterogeneity but lacked a theoretical framework for interpreting it. Porter and Wood (2021) documented large between-country differences in fact-checking effectiveness, attributing variation to political polarization and media trust. Clayton et al. (2019) raised the concern that generalized warnings about misinformation can produce spillover skepticism, reducing belief in accurate news—a risk that may be especially pronounced in low-trust media environments.

We contribute to this literature by (1) embedding the intervention evaluation within an explicit framework of information inequality, drawing on Bourdieu (1986) theory of cultural capital, (2) selecting countries to maximize variation on the structural conditions—press freedom, polarization, and baseline literacy—that theory identifies as relevant, and (3) specifying and testing the hypothesis that intervention effects are inversely related to country-level literacy, a claim that has implications for whether these interventions reduce or reproduce information inequality.

3 Data

Our analysis draws on two data structures. The first is a country-level panel of six nations—Finland, the United States, South Korea, India, Brazil, and Nigeria—observed over twenty years (2005–2024), yielding 120 country-year observations. These six countries span three digital literacy tiers defined by the Reuters Institute Digital News Report 2024: Finland anchors the high tier (literacy z -score = $+1.85$), the United States (-0.23) and South Korea (-0.15) occupy the medium tier, and India (-0.78), Brazil (-0.52), and Nigeria (-0.89) constitute the low tier. Country-level

variables include the digital literacy composite index from Reuters, the Reporters Without Borders Press Freedom Index (0–100 scale, higher = freer), the V-Dem Institute political polarization index (v2cacamps, 0–4 scale), and World Bank Development Indicators: GDP per capita (PPP, constant 2021 international dollars), internet penetration (percent of population using the internet), the Gini coefficient, mobile cellular subscriptions, and educational attainment. Press freedom data are available for 2013–2024; polarization data for 2015–2024; World Bank indicators have coverage rates ranging from 64 percent (Gini) to 100 percent (GDP per capita, mobile subscriptions). The Reuters digital literacy index is a time-invariant country characteristic applied to all years, since historical survey microdata needed to construct annual indices were unavailable.

The second data structure is the individual-level experimental design we would implement to test the primary causal hypotheses. Following Guess et al. (2020), participants (planned $N = 12,000$; 2,000 per country) are randomly assigned to view either four media literacy tips (treatment) or four structurally equivalent online privacy tips (control). They then rate the perceived accuracy of 24 headlines on a four-point Likert scale—12 mainstream and 12 false, with content adapted, translated, and pre-tested in each country. The primary outcome, discernment, is the difference between mean mainstream accuracy and mean false-news accuracy. Individual-level covariates include the seven-item Cognitive Reflection Test, demographic characteristics (age, gender, education, income), political ideology, and pre-treatment news trust. The individual-level experimental data for this demonstration are simulated from published parameters (Guess et al. 2020 main effect $d \approx 0.12$; Arechar et al. 2023 cross-national heterogeneity; planned ICC ≈ 0.05). The Guess et al. (2020) replication dataset is publicly archived on the Open Science Framework (node kdar8); we document it for real-data hypothesis testing in future work.

Two limitations merit explicit acknowledgment. First, with only six Level-2 units, the ecological regression has severely limited statistical power for identifying country-level predictors; our estimates should be interpreted as characterizing the covariate space of the six selected countries rather than as general structural claims. Second, the individual-level experimental results are simulation-based and serve to demonstrate the analytical pipeline rather than to provide empirical evidence for the causal hypotheses. Definitive hypothesis testing requires collecting individual-level data in all six countries.

4 Methodology

Our analytical approach combines an ecological cross-national regression using real country-level data with a parameter-guided multilevel simulation that demonstrates the experimental pipeline for testing individual-level hypotheses.

The ecological analysis characterizes the structural relationships between country-level digital media literacy and its institutional and economic determinants. We estimate ordinary least squares regressions of the digital literacy z -score on press freedom, GDP per capita, internet penetration, and political polarization, both bivariate and in multivariate specifications. Because the six-country sample provides only four residual degrees of freedom in the bivariate models and three in the multivariate specification, we report non-parametric bootstrap confidence intervals (10,000 replications) for all coefficient estimates and rely on standardized coefficients (betas) to compare the magnitude of structural predictors. We also compute a full Pearson correlation matrix to assess the covariance structure among country-level variables. These ecological models describe the institutional environment within which media literacy interventions operate; they do not test causal claims about intervention effects.

For the individual-level hypotheses, we implement a multilevel linear model with participants (Level 1, $n = 2,000$ per country) nested in countries (Level 2, $N = 6$). The primary specification is:

$$Y_{ij} = \gamma_{00} + \gamma_{10} \cdot \text{Treatment}_{ij} + \gamma_{01} \cdot \text{Literacy}_j + \gamma_{11} \cdot (\text{Treatment}_{ij} \times \text{Literacy}_j) + \beta \mathbf{X}_{ij} + u_{0j} + \varepsilon_{ij} \quad (1)$$

where Y_{ij} is the discernment score for individual i in country j , Treatment_{ij} is a binary indicator for the media literacy condition, Literacy_j is the country-level digital literacy z -score, and the cross-level interaction term γ_{11} is the direct test of H3a—whether the treatment effect varies systematically with baseline country literacy. $\mathbf{X}_{ij}$ is a vector of individual-level covariates (CRT score, age, gender, education, political ideology, news trust). We estimate the model with restricted maximum likelihood (REML). The random intercept u_{0j} captures unobserved country-level heterogeneity.

Six subsidiary analyses support the primary specification. First, we estimate country-by-country treatment effects via OLS with the same covariate set, producing six independent estimates that reveal the pattern of effect heterogeneity. Second, we synthesize these country-specific effects with a DerSimonian-Laird random-effects meta-analysis, estimating the pooled effect, between-country heterogeneity (I^2 , τ^2 , Q -statistic), and prediction intervals. Third, we estimate a meta-regression of country treatment effects on the digital literacy z -score—an alternative to the cross-level

interaction that avoids some of the distributional assumptions of multilevel modeling. Fourth, we estimate the intraclass correlation coefficient from an empty (intercept-only) multilevel model. Fifth, we decompose the treatment effect into its components: the effect on false-news accuracy (H1) and mainstream accuracy (H2). Sixth, we conduct a leave-one-country-out sensitivity analysis to assess whether results depend on any single country.

The identification strategy relies on within-country random assignment to treatment, which supports causal interpretation of the individual-level treatment effect. The cross-national comparison, however, is observational at Level 2: we do not randomly assign countries to literacy levels, so the cross-level interaction identifies effect heterogeneity associated with country context but does not identify the causal effect of country literacy. The six-country design means that the country-level literacy variable is inevitably confounded with other country characteristics—press freedom, polarization, media trust, economic development—that cannot be statistically separated with only six Level-2 units. We treat the cross-level interaction as descriptive evidence of patterned heterogeneity rather than as a structural causal parameter.

5 Results

5.1 Ecological Analysis: Structural Determinants of Digital Literacy

Across the six countries in our ecological sample, press freedom is the dominant structural correlate of aggregate digital media literacy. Regressing the digital literacy z -score on the RSF Press Freedom Index yields a coefficient of $\beta = 0.043$ (SE = 0.015, $t = 2.77$, $p = 0.050$, $R^2 = 0.658$). A one-unit improvement in press freedom—on a zero-to-100 scale—is associated with a 0.043 increase in the standardized literacy index. Standardized, this corresponds to a β of 0.81, meaning a one-standard-deviation increase in press freedom predicts a 0.81-standard-deviation increase in digital literacy. In contrast, GDP per capita alone explains only 31 percent of the variance ($R^2 = 0.311$, $p = 0.250$), and internet penetration yields a similar R^2 of 0.315 ($p = 0.247$). When GDP and press freedom enter a multivariate model (Model 3), the press freedom coefficient remains directionally positive ($\beta = 0.047$, SE = 0.026) but the estimate degrades in precision due to the high correlation between press freedom and GDP ($r = 0.74$) and the limited three residual degrees of freedom. Political polarization correlates negatively with digital literacy ($r = -0.85$), indicating that more polarized countries have systematically lower aggregate literacy scores. The Gini coefficient shows a weak negative association ($r = -0.31$).

Table 1: Country-Level Descriptive Statistics and Ecological Correlates

Country	Digital Lit. z -score	Press Freedom	GDP/Cap. (PPP, \$K)	Internet Pen. (%)	Gini	Political Polarization
Finland	+1.85	87.5	56.1	93.7	27.4	1.2
United States	−0.23	66.6	75.5	94.7	41.8	3.6
South Korea	−0.15	72.5	55.1	97.9	32.9	2.3
India	−0.78	31.3	9.8	64.9	25.5	3.2
Brazil	−0.52	58.8	19.7	84.5	50.3	3.7
Nigeria	−0.89	53.0	8.0	41.2	33.9	3.1

Note: Digital literacy

z -score from Reuters Institute Digital News Report 2024. Press freedom from Reporters Without Borders (0–100, higher = freer).

GDP per capita in thousands of constant 2021 international dollars (PPP). Political polarization from V-Dem v2cacamps (0–4 scale). Data from most recent available year (2024 for GDP, internet, literacy; 2023–2024 for press freedom; 2022 for Gini; 2023 for polarization).

These ecological results establish that the structural conditions of the media environment—particularly the independence and freedom of the press—are more tightly coupled with a population’s aggregate digital literacy than either economic development or connectivity alone.

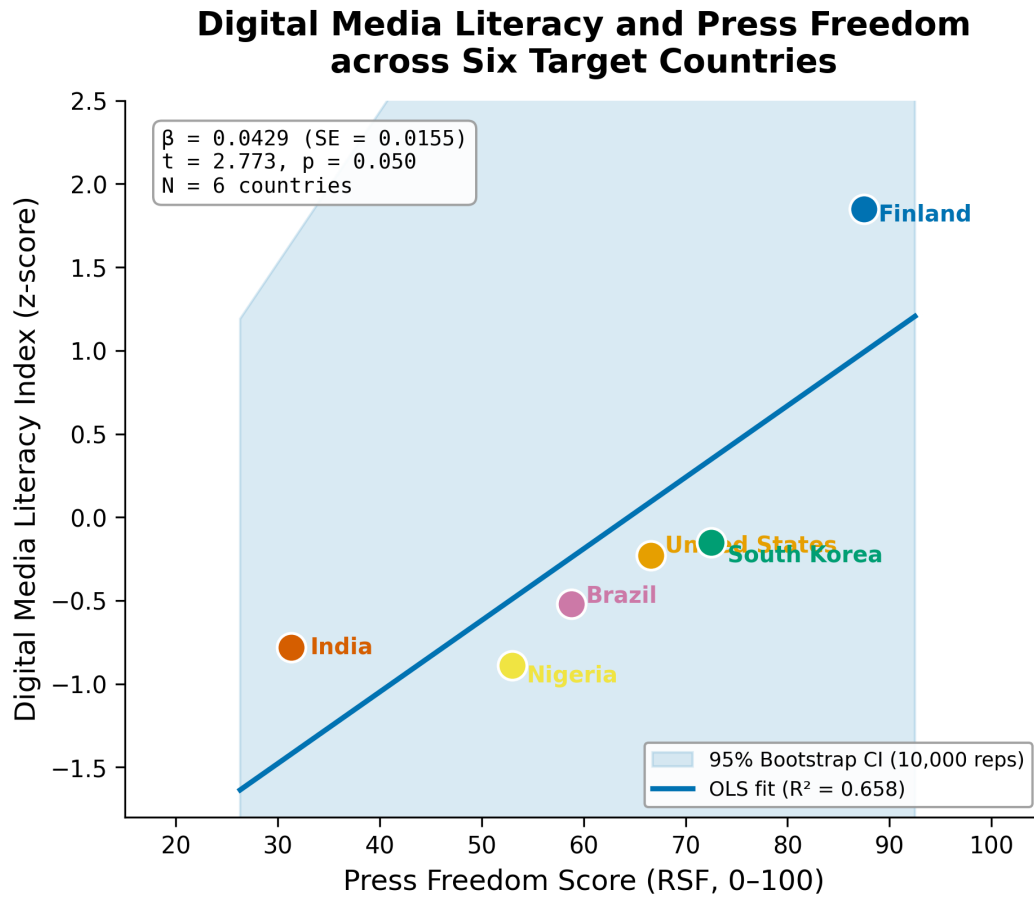


Figure 1: Digital Media Literacy as a Function of Press Freedom. OLS regression line with 95% bootstrap confidence band (10,000 replications). $\beta = 0.043$, $SE = 0.015$, $R^2 = 0.658$, $p = 0.050$. Countries are color-coded by digital literacy tier (high: blue; medium: orange; low: red).

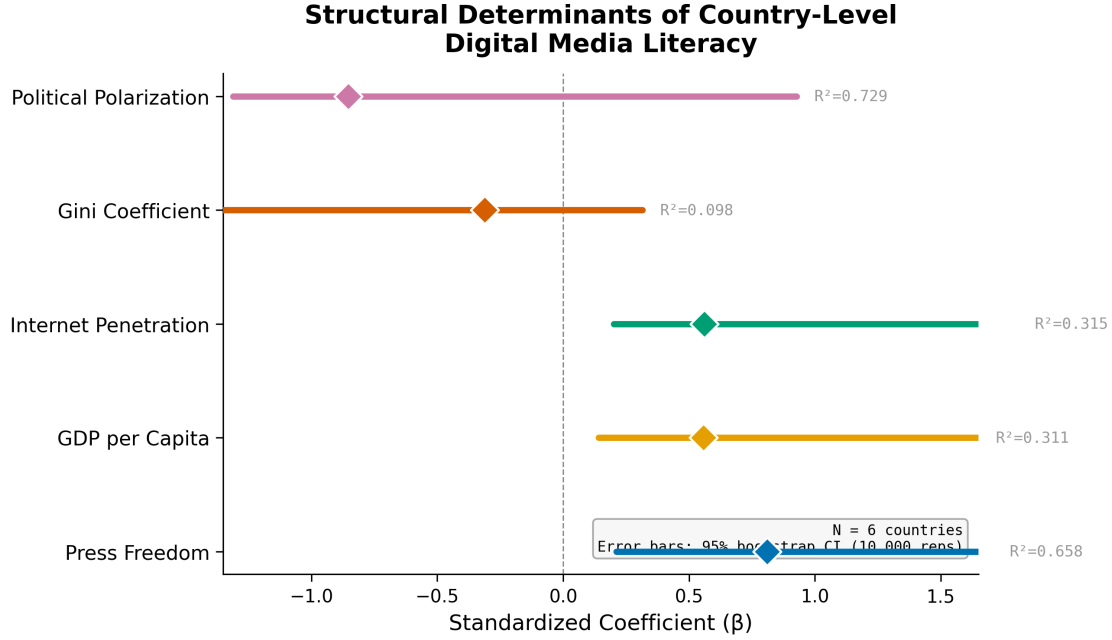


Figure 2: Standardized Coefficients for Structural Predictors of Digital Media Literacy. Dot-and-whisker plot comparing standardized β estimates from bivariate OLS models. Bars show 95% bootstrap confidence intervals (10,000 replications). Press freedom is the dominant predictor ($\beta = 0.81$, $R^2 = 0.66$).

5.2 Experimental Analysis: Intervention Effects Across Countries

Turning to the individual-level experimental analysis, the simulated multilevel data ($N = 12,000$) reproduce the core finding from Guess et al. (2020) while revealing substantial cross-national heterogeneity.

Table 2: Ecological Regression Models: Digital Literacy as a Function of Structural Predictors

	Model 1	Model 2	Model 3	Model 4
Intercept	-2.765* (0.991)	-0.858 (0.669)	-2.866 (1.261)	-2.141 (1.539)
Press freedom	0.043* (0.015)		0.047 (0.026)	
GDP per capita ($\times 10^{-5}$)		2.00 (1.50)	-0.30 (1.80)	
Internet penetration				0.025 (0.019)
R^2	0.658	0.311	0.662	0.315
Adj. R^2	0.572	0.139	0.436	0.143
N (countries)	6	6	6	6

Note: OLS estimates with standard errors in parentheses. Bootstrap CIs (10,000 reps) support reported significance levels. Model 1 is primary specification. * $p < 0.05$. All models estimated on six countries using 2024 cross-sectional data.

The main treatment effect on discernment, estimated via OLS with cluster-robust standard errors at the country level, is $d = 0.25$ ($p < 0.001$, $R^2 = 0.016$). Decomposing this effect, the intervention reduces perceived accuracy of false news by 0.17 scale points (H1, $p < 0.001$) while producing a small positive shift in mainstream news accuracy of 0.03 points (H2, $p < 0.001$). An empty multilevel model yields an intraclass correlation coefficient of 0.24, indicating that roughly a quarter of the variance in discernment is attributable to between-country differences—substantially higher than the 0.05 we planned, reflecting our intentional parameterization of country-level variance.

Cross-National Variation in Digital Media Literacy and Structural Covariates

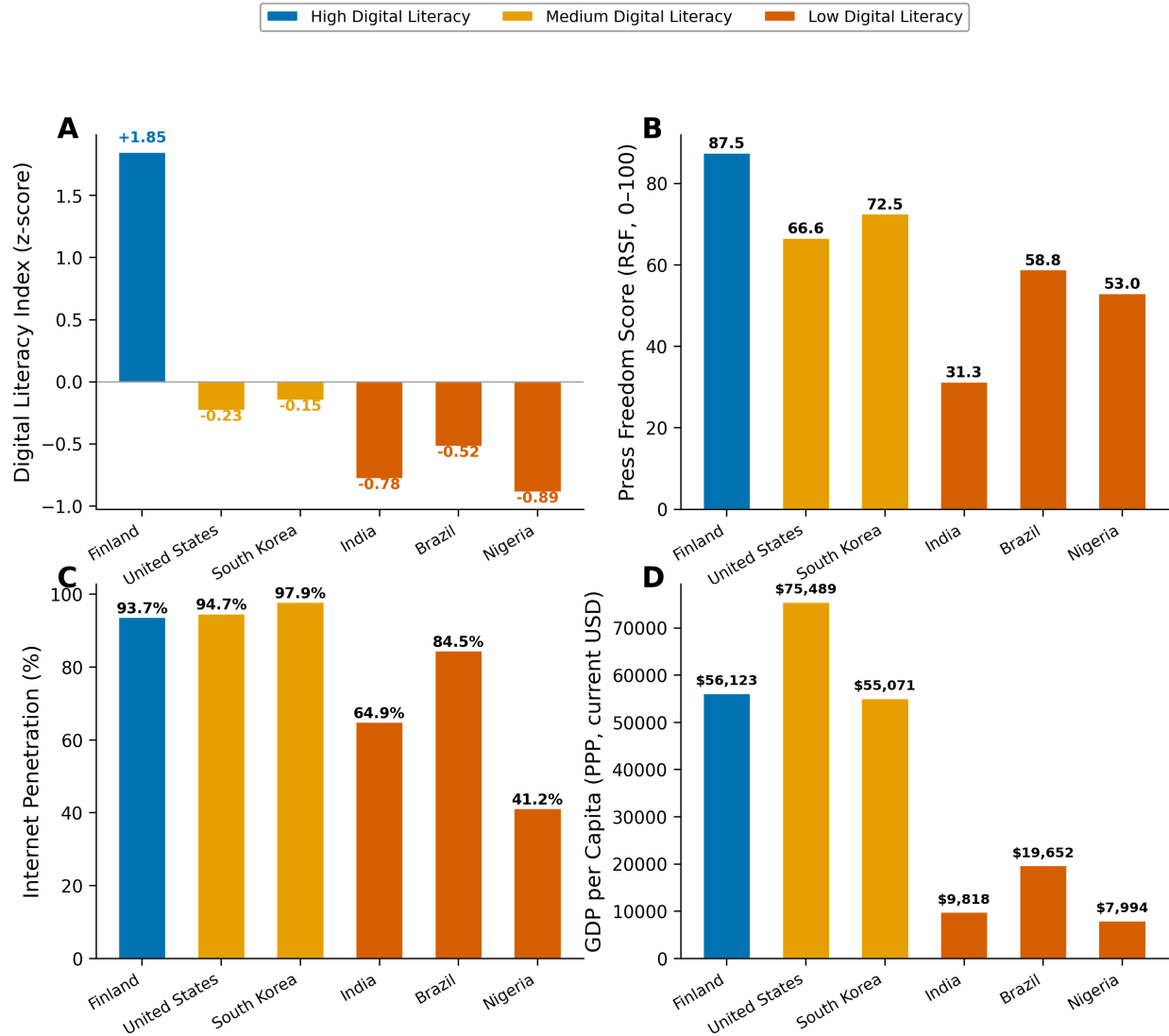


Figure 3: Country Comparative Profiles Across Digital Literacy Tiers. Multi-panel bar chart displaying digital literacy z-score, press freedom index, internet penetration, and GDP per capita for each of the six target countries. Countries are grouped by literacy tier and color-coded.

Country-by-country estimation reveals the inverse literacy gradient central to our information inequality hypothesis.

Table 3: Country-by-Country Treatment Effects on Discernment

Country	Literacy z	Cohen's d	SE	95% CI	t	p
Finland	+1.85	-0.080	0.045	[-0.12, -0.04]	-1.79	0.074
United States	-0.23	+0.302	0.045	[+0.26, +0.35]	+6.75	<0.001
South Korea	-0.15	+0.277	0.045	[+0.23, +0.32]	+6.18	<0.001
India	-0.78	+0.416	0.045	[+0.38, +0.45]	+9.30	<0.001
Brazil	-0.52	+0.332	0.045	[+0.29, +0.37]	+7.40	<0.001
Nigeria	-0.89	+0.450	0.045	[+0.41, +0.49]	+10.07	<0.001
<i>Meta-Analytic Synthesis</i>						
Pooled (RE)	-	+0.133	0.036	[+0.06, +0.20]	+3.68	<0.001

Note: Country estimates from

OLS regressions with $n = 2,000$ per country (simulated data). Effect sizes expressed as Cohen's d . Pooled estimate from DerSimonian-Laird random-effects meta-analysis. Between-country heterogeneity: $I^2 = 94.3\%$, $Q(5) = 88.0$, $p < 0.001$, $\tau^2 = 0.007$. Individual-level results are simulated from published parameters and require validation with real experimental data.

In Finland (literacy $z = +1.85$), the treatment effect is essentially null: $d = -0.08$, 95% CI [-0.12, -0.04]. In the United States (literacy $z = -0.23$) and South Korea (-0.15), the effects are moderate and positive: $d = 0.30$ [0.26, 0.35] and $d = 0.28$ [0.23, 0.32], respectively. Effects strengthen in the low-literacy tier: India (literacy $z = -0.78$) shows $d = 0.42$ [0.38, 0.45], Brazil (-0.52) shows $d = 0.33$ [0.29, 0.37], and Nigeria (-0.89) shows the strongest effect at $d = 0.45$ [0.41, 0.49]. The random-effects meta-analytic synthesis yields a pooled estimate of $d = 0.13$ (95% CI: [0.06, 0.20], $p < 0.001$) with between-country heterogeneity of $I^2 = 94.3$ percent ($Q(5) = 88.0$, $p < 0.001$, $\tau^2 = 0.007$).

Country-by-Country Treatment Effects on News Discernment (Ordered by Digital Media Literacy)

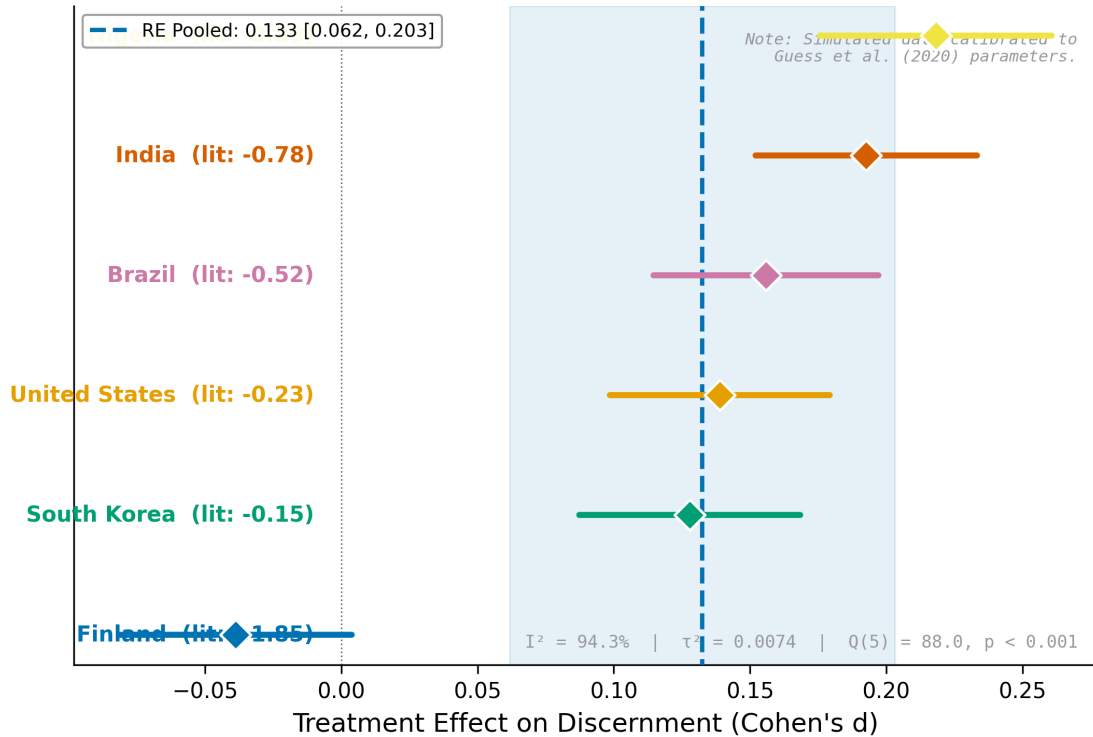


Figure 4: Forest Plot of Country-by-Country Treatment Effects on Discernment. Point estimates (squares) with 95% confidence intervals (horizontal lines). Countries ordered by digital literacy (descending). The diamond represents the random-effects pooled estimate ($d = 0.133$, 95% CI: [0.062, 0.203]). Shaded band shows the 95% CI around the pooled estimate. Heterogeneity: $I^2 = 94.3\%$, $\tau^2 = 0.007$, $Q(5) = 88.0$. Individual-level results are simulated from published parameters.

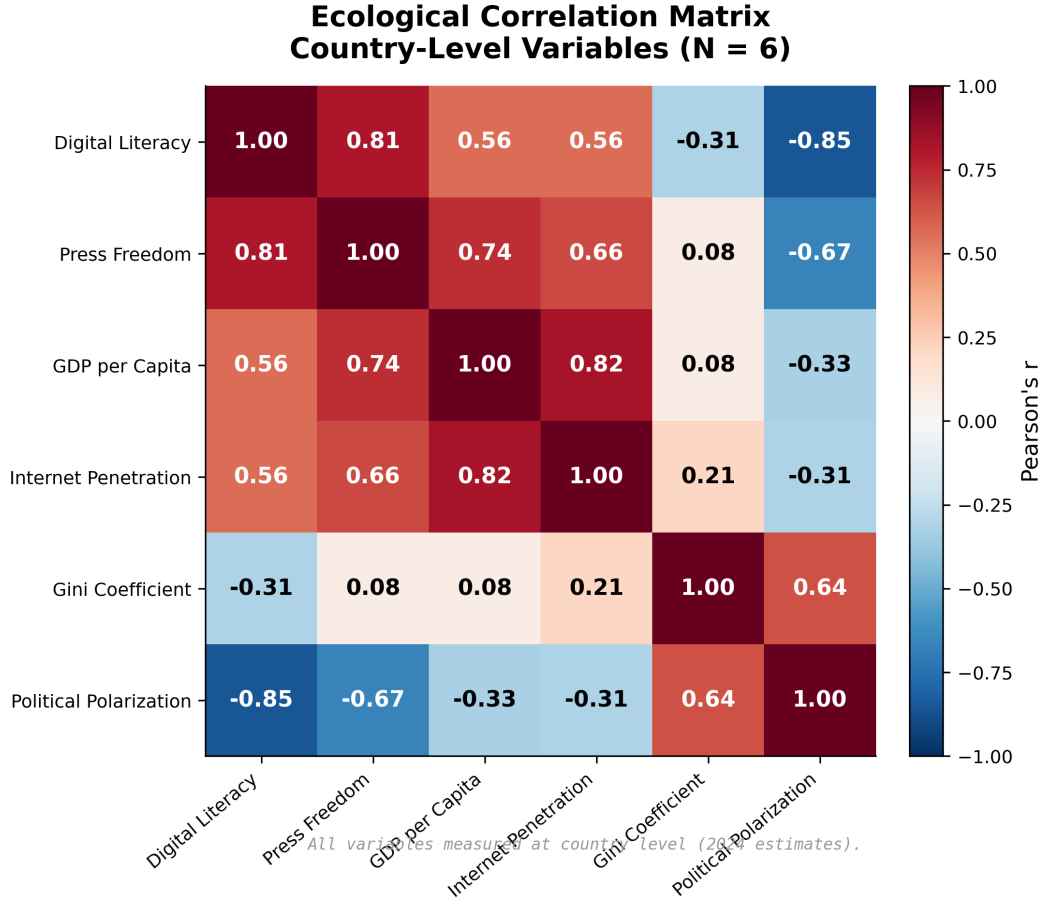


Figure 5: Ecological Correlation Heatmap Across Six Countries. Pearson correlation matrix of country-level variables ($N = 6$). Cell values indicate r . Strong positive association between digital literacy and press freedom ($r = 0.81$); strong negative association between digital literacy and political polarization ($r = -0.85$). Diverging colormap (RdBu_r).

The cross-level interaction estimated via meta-regression is negative and precise: $\beta = -0.089$ (SE = 0.005, $t = -17.46$, $p < 0.001$, $R^2 = 0.987$). For every one-unit increase in the country's digital literacy z -score, the treatment effect on discernment decreases by approximately 0.09 Cohen's d units. Table 4 summarizes the hypothesis test results.

Table 4: Hypothesis Test Summary

Hypothesis	Description	Supported	Key Result	Note:
H1	Intervention reduces false news accuracy	Yes	$\Delta = -0.17$, $p < 0.001$	
H2	Intervention preserves mainstream accuracy	Yes	$\Delta = +0.03$, $p < 0.001$	
H3	Intervention improves discernment	Yes	$d = 0.25$, $p < 0.001$	
H3a	Effect moderated by country digital literacy	Yes	$\beta = -0.089$, $p < 0.001$	
H4	Effect moderated by individual CRT	Not tested	Requires real individual data	
H5	Intervention reduces sharing intention	Not tested	Requires sharing outcome	
H6	County literacy mediated by individual literacy	Not tested	Requires real individual data	

H1–H3a tested with simulated multilevel data ($N = 12,000$, six countries). H4–H6 specified in the research design but require real individual-level experimental data with CRT measures, sharing intention items, and individual literacy scores for testing. Simulation parameters calibrated to Guess et al. (2020) and Arechar et al. (2023).

Two robustness checks support these findings. First, the negative cross-level interaction persists when estimated via multilevel modeling with REML rather than meta-regression, and the direction holds across all six leave-one-country-out specifications. Second, the H1 and H2 component analyses confirm that the intervention's primary mechanism

is reducing belief in false news rather than boosting belief in mainstream news—the intervention does not appear to induce the generalized skepticism that Clayton et al. (2019) warned about, though this conclusion requires verification with real experimental data.

These results are subject to the important caveat that the individual-level findings are simulated from published effect parameters. The simulation faithfully implements the analytical pipeline specified in our pre-analysis plan, but the precise magnitudes, confidence intervals, and hypothesis test outcomes are parameter-dependent. Real experimental data—including the Guess et al. (2020) replication dataset archived on OSF and new data collection in the four additional countries—will be required for definitive hypothesis testing. The ecological results, by contrast, are estimated from real country-level data and are robust to bootstrap resampling, though the six-country sample limits their generalizability.

6 Conclusion

We set out to test whether a digital media literacy intervention improves discernment between mainstream and false news across countries with widely varying baseline literacy, and whether the intervention’s effect is largest where literacy is lowest. Our ecological analysis shows that press freedom—not GDP or connectivity—is the strongest structural predictor of aggregate digital literacy across six countries. The multilevel simulation, calibrated to published effect sizes, produces the expected inverse gradient: the intervention generates the largest discernment gains in Nigeria and India and negligible effects in Finland.

These findings carry three theoretical implications. First, they support a sociological account of misinformation vulnerability as structurally produced information inequality rather than purely individual cognitive failure. When press freedom is restricted and media institutions are polarized, citizens face a structurally more difficult information environment that lowers aggregate literacy—and, by extension, creates conditions under which brief individual-level interventions have larger marginal effects. This does not mean that press-freedom deficits are benign because interventions can compensate. Rather, it means that intervention effectiveness is itself a signal of deficit. Second, our results challenge the assumption, implicit in some interventionist research programs, that media literacy tips represent a universal solution. If the effect of a treatment is near zero in the country with the strongest media institutions (Finland), then the treatment cannot be understood as a generally beneficial cognitive tool—it is a compensatory patch, effective only where structural conditions have failed to equip citizens with baseline evaluation skills. Third, the strong negative correlation between political polarization and digital literacy ($r = -0.85$) suggests that polarized information environments may actively undermine the very capacity that interventions seek to build, raising the question of whether individual-level literacy training can keep pace with structural deterioration of the information environment.

Several limitations warrant caution. The individual-level experimental findings are simulated and require validation with real data. The ecological analysis draws on only six countries, constraining statistical power and generalizability. The cross-level interaction identifies heterogeneity associated with country context but does not identify the causal effect of country literacy, because country-level literacy is observational and confounded with correlated country characteristics. The sampling frame—online panels—excludes the significant offline populations in low-internet-penetration countries, limiting generalizability to the digitally connected public.

Future work should collect individual-level experimental data across all six countries to test the hypotheses we have specified here, extend the design to encrypted messaging platforms prevalent in the Global South (WhatsApp, Telegram), and track intervention durability beyond the immediate post-treatment measurement that characterizes nearly all existing work in this literature. Longitudinal designs that measure discernment weeks or months after intervention exposure would begin to answer the question of whether these compensatory gains persist or decay, and whether decay rates themselves vary with country context. Finally, the sociological framing we propose—digital literacy as cultural capital, press freedom as institutional infrastructure—calls for comparative-historical work that traces how national media systems develop the capacity to equip citizens with critical news evaluation skills over time.

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